IMEN574 / IDSC574 · PROGRAMMING FOR AX

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Language Models

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Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

Outline: Language Models

01 **Contextual Tokens**

How tokens read context

02 **Training and Generation**

How objectives shape outputs

03 **Adapting and Verifying**

How evidence enters answers

01 · Contextual Tokens

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Why Model Sequences

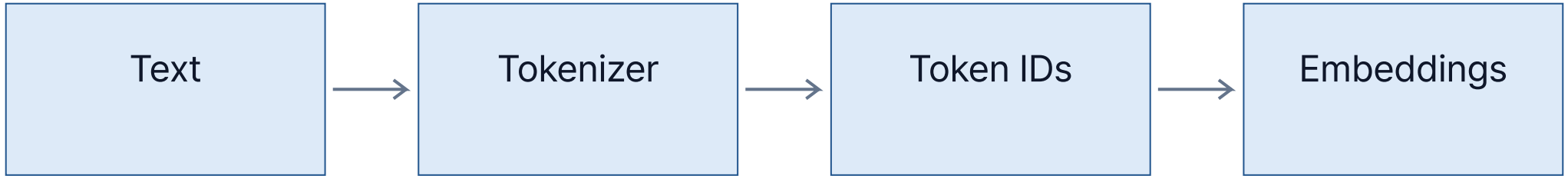
The same token can mean different things in context

Context A	river	bank
Context B	savings	bank

Context changes the information carried by a token

Tokens and Vocabulary

The model predicts token IDs rather than whole ideas



Text becomes an indexed sequence

Embedding Lookup

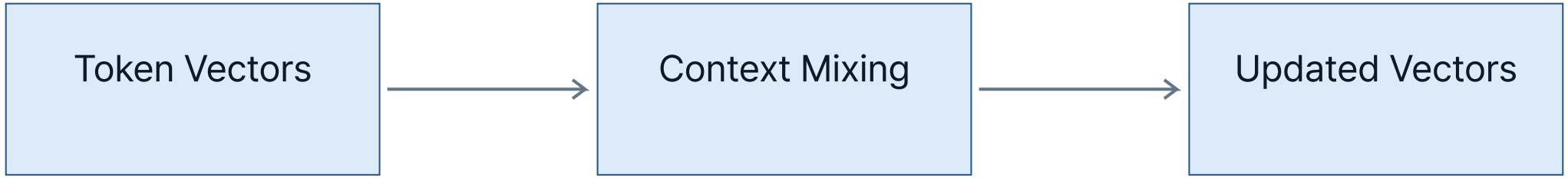
Token IDs select trainable vectors

	d1	d2
ID 0	0.2	0.4
ID 1	0.7	-0.1
ID 2	-0.3	0.9

An ID selects a row in the embedding table

Contextual Representations

A token vector changes after it reads other tokens

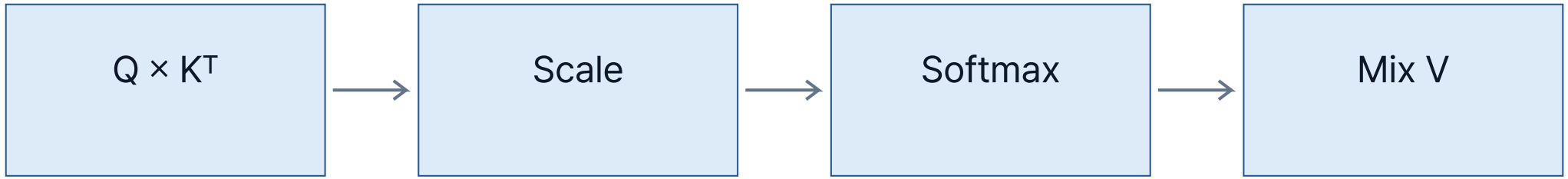


Context mixing changes the vector at each position

Scaled Dot-Product Attention

Queries choose a weighted mixture of values

$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T / \sqrt{d_k})V$$



Queries, Keys and Values

One input can supply three learned projections

	Query	Key	Value
Token 1	q1	k1	v1
Token 2	q2	k2	v2

Self-attention projects one sequence three ways

Attention Scores

Each query has a distribution across key positions

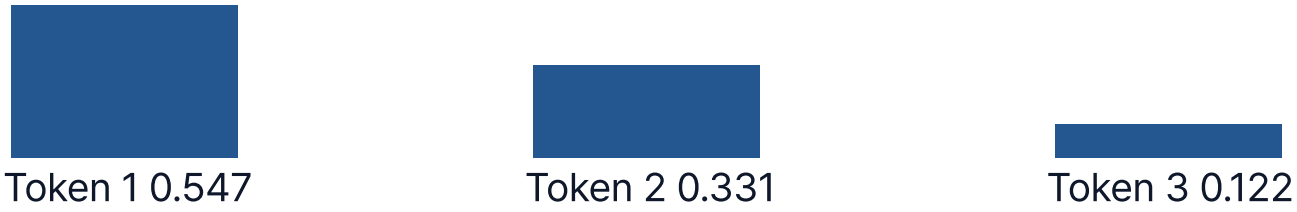
	key 1	key 2	key 3
query 1	0.7	0.2	0.1
query 2	0.1	0.8	0.1
query 3	0.2	0.3	0.5

Each row defines its own weighted mixture

Demo: Attention Temperature

Change temperature and inspect the weighted value

Temperature 1

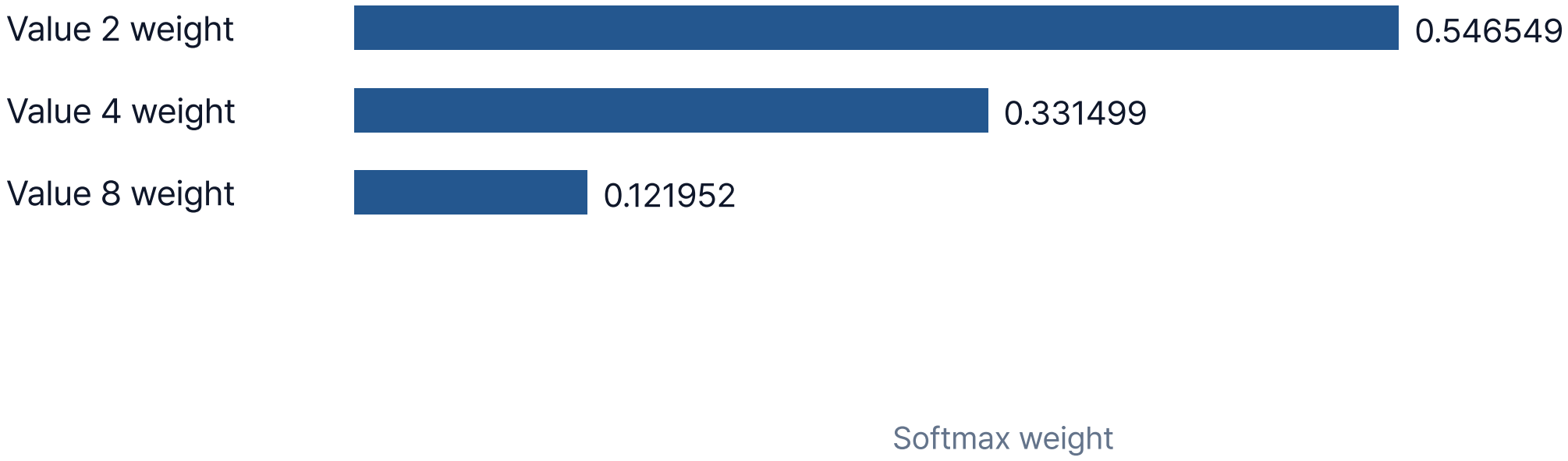


softmax([1, 0.5, −0.5] / T) | T 1.00

Weighted value 3.395 for values [2, 4, 8]

Answer: Weighted Values

Temperature one produces a convex combination



The weighted output is approximately 3.395

Attention Masks

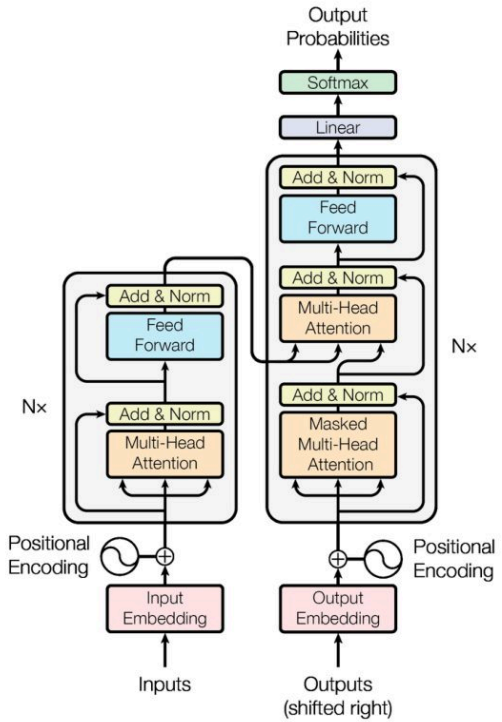
A mask excludes disallowed information paths

	key 1	key 2	key 3
position 1	1	0	0
position 2	1	1	0
position 3	1	1	1

Causal masking blocks future token access

Transformer Block

Attention is one component inside a residual block



The block combines mixing and per-token transformations

https://arxiv.org/abs/1706.03762

02 · Training and Generation

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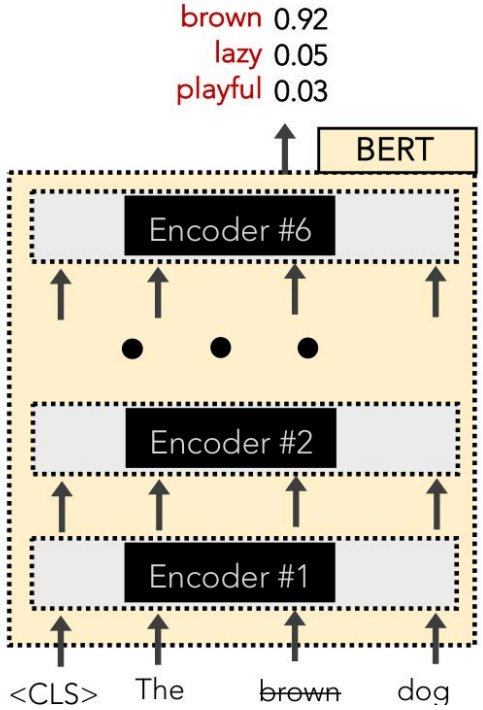
Why Two Training Objectives

Reading all tokens and predicting the next token need different masks

Objective	Visible context	Use
Masked token	Both directions	Representation
Next token	Prefix only	Generation

Masked Language Modeling

The model predicts hidden tokens from surrounding context

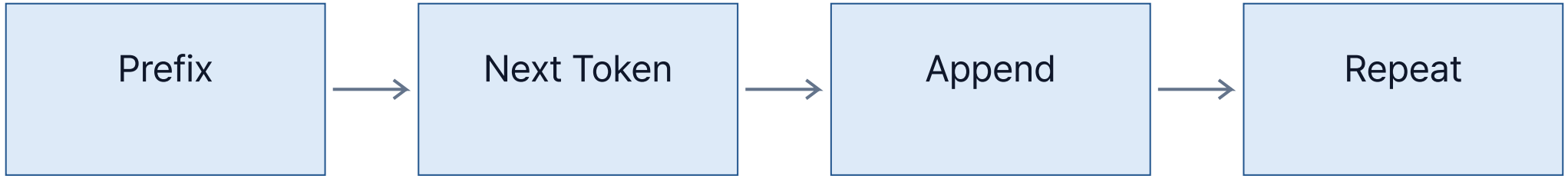


Both sides can inform a masked position

https://arxiv.org/abs/1810.04805

Autoregressive Factorization

A sequence probability is a product of conditional predictions



Each generated token becomes part of the next context

Next-Token Loss

Training compares each prefix with the observed next token

$$L = -\Sigma_t \log p(x_t \mid x_{<t})$$

Shifted Inputs and Targets

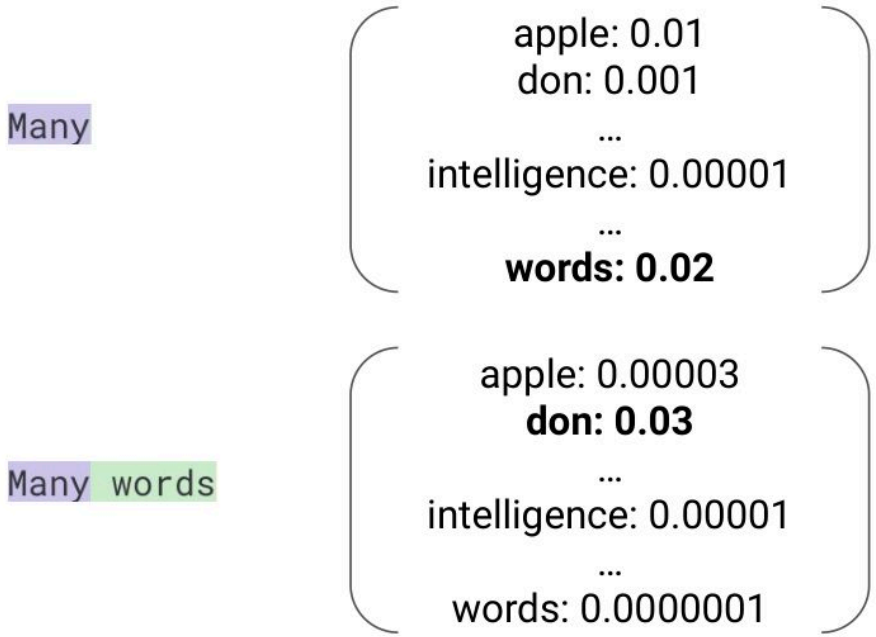
Every observed transition supplies a training example

	t1	t2	t3
Inputs	The	cat	sat
Targets	cat	sat	EOS

Targets are offset by one position

Token Generation

The next-token distribution grows one prefix at a time



A sampled token extends the prefix

https://arxiv.org/abs/2005.14165

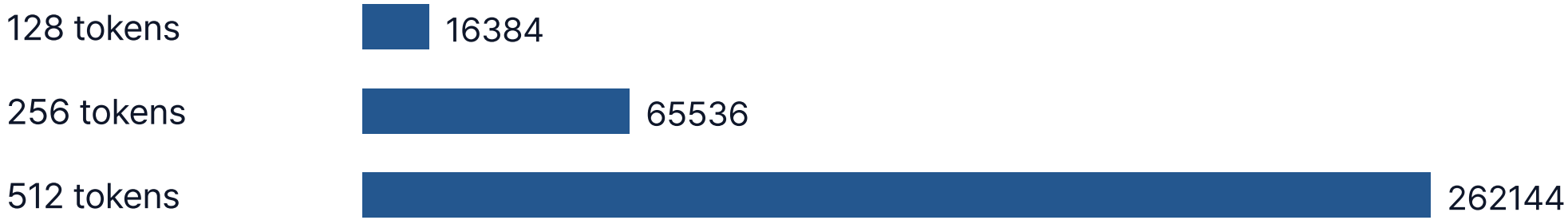
Decoding Policy

The same logits can produce different output sequences

Policy	Rule
Greedy	Choose the largest logit
Sampling	Draw from probabilities
Top-p	Restrict cumulative mass

Context Length and Cost

Longer context expands the token interaction matrix

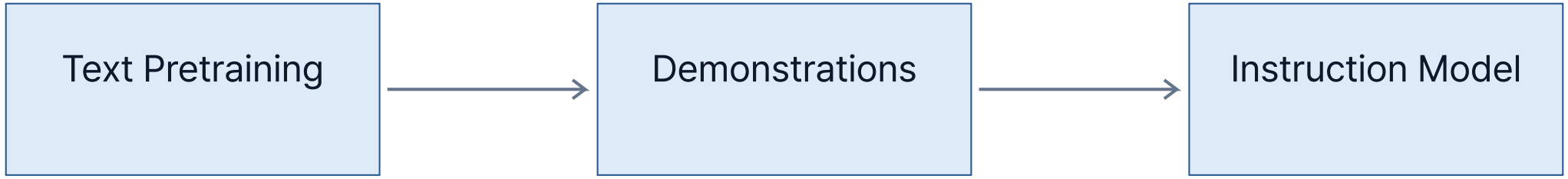


Dense attention score entries

Doubling sequence length quadruples score entries

Pretraining and Instruction Tuning

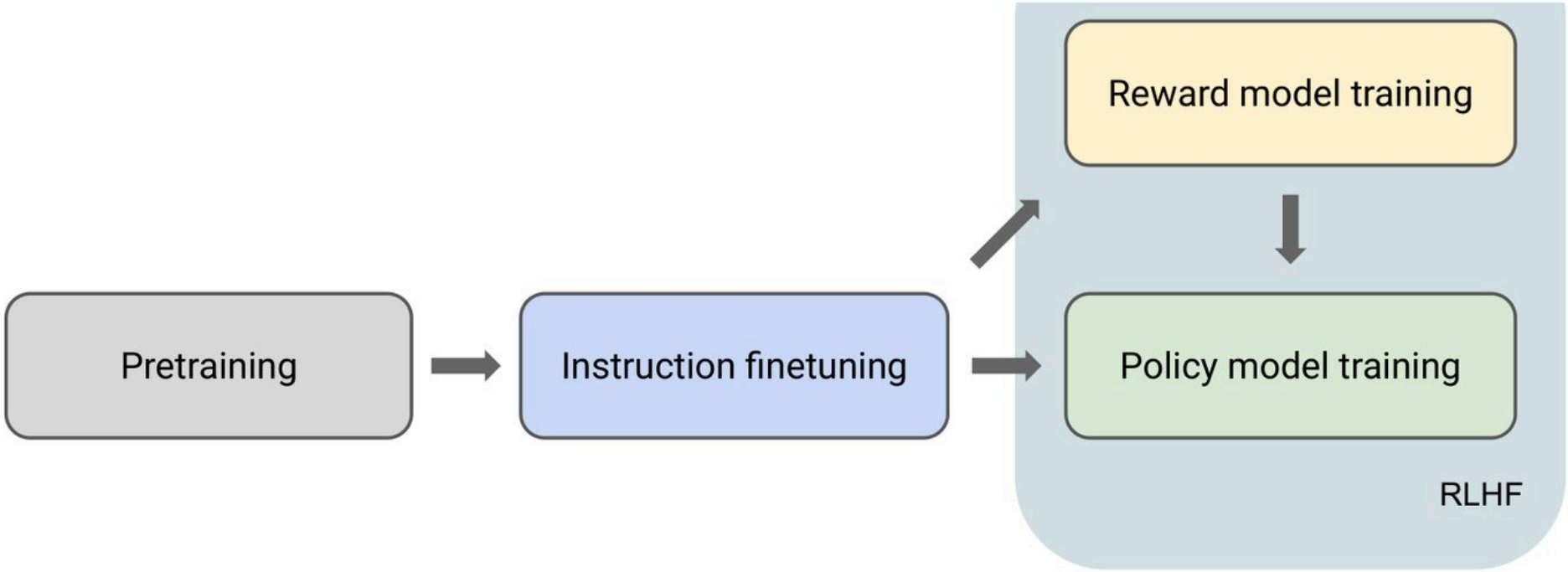
Following a request is trained after learning broad text patterns



Demonstrations teach the model a response format

Human Feedback Pipeline

Preference signals can further shape response behavior



Preference training changes which responses are favored

https://arxiv.org/abs/2203.02155

Caveat: Fluent Errors

Language likelihood does not certify factual truth

Item	Field 1	Field 2
Surface	Plausible wording	high
Evidence	Source support	absent

A likely continuation can lack supporting evidence

03 · Adapting and Verifying

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Why Adaptation Methods Differ

A model may need knowledge, behavior or examples

Need	Mechanism
New document evidence	Retrieval
A response format	Context examples
Persistent adaptation	Parameter updates

In-Context Learning

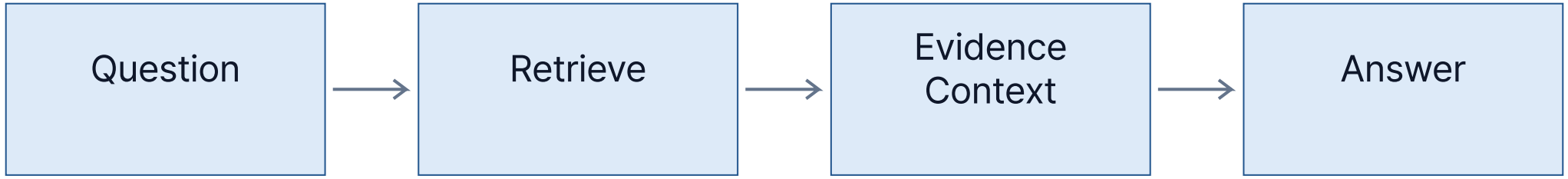
Examples change the context while weights stay fixed

Item	Field 1	Field 2
Examples	great → positive	poor → negative
Query	fine → ?	weights unchanged

Context examples change the conditional prediction

Retrieval-Augmented Generation

Retrieved text supplies evidence for a current question



Evidence enters through the prompt context

Retrieval Ranking

The chosen documents determine the available evidence



Illustrative retrieval similarity

High similarity is a ranking signal, not proof

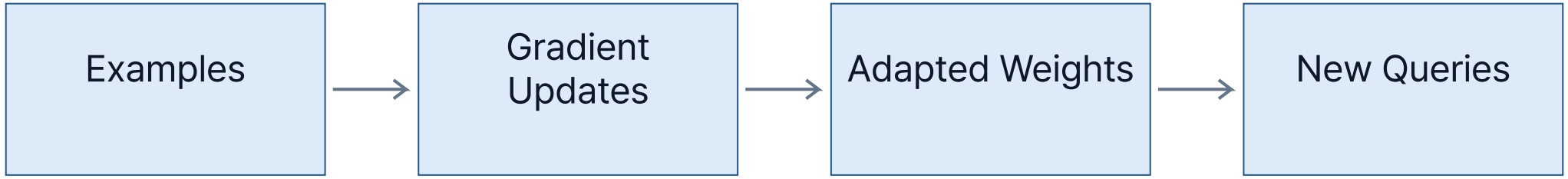
Grounded Answer Contract

An answer must retain links to the supporting passages

Claim	Evidence ID
Valve opened at 10:05	Log 17
Pressure fell later	Sensor 9

Fine-Tuning Scope

Parameter updates persist beyond a single prompt



Fine-tuning changes the stored model parameters

Structured Output Validation

Parseable output still needs semantic checks

```
answer = {"count": 3, "unit": "items"}
assert isinstance(answer["count"], int)
assert answer["count"] >= 0
assert answer["unit"] == "items"
```

Prompt Evaluation Set

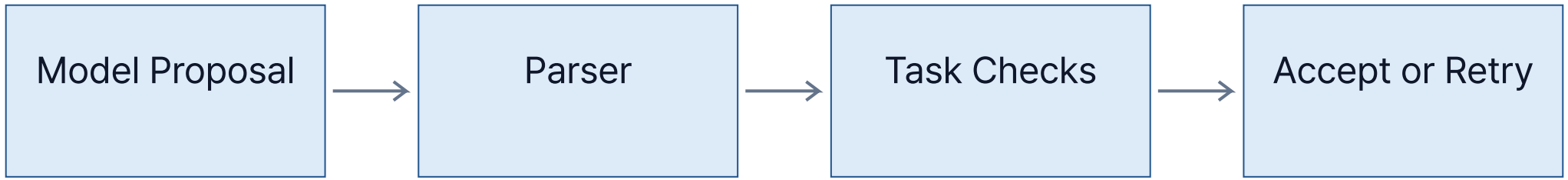
Keep examples that expose known failure modes

	Prompt A	Prompt B	Prompt C
Simple	1	1	0
Ambiguous	1	0	1
Adversarial	0	1	1

Different prompts fail on different input slices

Model and Harness Boundary

Generation supplies a proposal while software checks the result



A software check decides whether to use a proposal

Language Model Audit

Record context, weights and decoding as separate choices

Component	Record
Context	Prompt and retrieved passages
Weights	Model and adaptation
Decoding	Sampling and stop rules

Takeaway: Language Models

01 A language model predicts conditioned token sequences

02 Context, weights and decoding are different controls

03 Fluent output still requires task-specific evidence